SUBJECT: COMPUTER SCIENCE / ARTIFICIAL INTELLIGENCE

Local AI Systems: Neural Architectures & Edge Runtime

Prepared for Offline AI Knowledge Ingestion & RAG Verification

1. Executive Summary

Local artificial intelligence decouples neural computation from remote cloud servers. By executing models on local silicon using quantized weight formats and lightweight runtimes, edge systems guarantee zero telemetry, microsecond network latency, and complete user privacy.

2. Component Pipeline

- **Neural Text-to-Speech (TTS):** Kokoro-82M ONNX model operating at 24kHz float32 audio. Delivers 54 distinct multilingual voice styles with sub-100ms first-packet generation.
- **Speech Recognition (STT):** Faster-Whisper small INT8 quantization with CPU-level Voice Activity Detection (VAD) and token timestamps.
- **Dual-Brain Language Model:** Dynamic switching between Qwen3-0.6B (sub-1.5s real-time voice conversations) and Qwen3-1.7B (deep multi-step coding, mathematical reasoning, and document analysis).
- **Local Vector RAG:** In-memory TF-IDF and cosine similarity embeddings chunking local documents into 350-word passages with source attribution.

3. Comparative Latency & Security Analysis

Centralized cloud APIs exhibit typical round-trip latencies between 600ms and 2500ms due to TLS negotiation, payload serialization, and queueing. In contrast, local inference maintains zero transport overhead, eliminating man-in-the-middle vulnerabilities and subscription costs.